

Section 2.5: Control Devices

In this unit we have discussed both input and output devices, including the different ways data is put into computers and how the resulting information is sent out.

The last topic covered in this unit is '**control devices**'. You will see how these are **used in conjunction with sensors** to '**make something happen**'.

Key Concepts of this section:

- # Know what the **main control devices** are.
- # Understand the **role of sensors** and **actuators** within control devices.
- # Be able to **discuss uses** of different types of control devices.

Control Devices (what are they?)

Definition: "Output devices that work in conjunction with sensor input devices and commands from computers in order to control a process."



Key Words:

Input, Process, Output, Sensor, Control Device, Actuator

Overview of Control Devices:

- # Control devices are **output devices** which **work together** with sensors.

Remember - Sensors gather '**measurable**' data from the environment and input it into a computer. They do not need any human involvement.

Sensors typically gather data such as **temperature, pressure, pH** etc.

- # Data gathered by a sensor is **input into a computer** which then decides what to do with the data (processes it).
- # Once the computer has decided what to do with the data it **sends instructions** to the control device.
- # The control device then **physically changes something** in response to commands received from the computer.

Consider this example:

Examples:



INPUT - Burglar breaks light sensor and this data is sent to the computer.



INPUT - A burglar breaks the **infrared light sensor** on a security system.

PROCESS - This data is sent to a **small computer** inside a security light.

The computer sends instructions to an '**actuator**' telling it to switch on the security light.

OUTPUT - The actuator **switches on** the security light.

Note: See below for a description of what an actuator is.



PROCESS - Computer looks at the data and instructs an actuator to switch on the security light.



OUTPUT - The actuator switches on the security light and the burglar is scared away.

NOTE:

Sensors and control devices are discussed in more detail in a later unit (link [here](#)).

Actuators ---- what are they?

- # Actuators take **instructions from the computer** and then make the '**physical**' part of the process happen.

For example - In our security system example above, the computer has decided that it is necessary to switch on the security light but it **cannot** do this by itself.

Instead the **computer instructs the actuator** to **switch the light on**.

- # Think of actuators as kind of like **muscles**.

We cannot move our limbs without using our muscles in much the same way as a light in a control device cannot turn on without an actuator.

Actuators respond to instructions sent by a **computer**.

Our muscles respond to instructions sent by our **brain**.

- # Actuators can also physically operate devices such as **motors, pumps, switches** and **valves**.



Examples of Control Devices



Key Words:

Motor, Buzzer, Heater, Lights

- # The table below shows you some examples of control devices:

Motor	Buzzer	Heater	Light / Lamp	REMEMBER! Control devices require inputs from sensors and instructions from a computer in order to produce the correct output.



Uses of Control Devices

The table below summarises the main uses of different control devices:

Remember - each of these control devices are switched on and off by an actuator

Type of Sensor:	Used In:
Motor	Households: • Automatic washing machines (to make drum spin) • DVD player (to make disc spin) • Air conditioners (to make fans spin) Industry: • Controlling robot arms Computer systems: • Spinning disks in hard disk drives • Spinning cooling fans
Buzzer	• Microwave ovens (to signal that the food is cooked) • Alarm systems (loud siren to warn of intruders)
Heater	• Central heating systems (to keep rooms correct temperature) • Washing machines (To keep water the correct temperature) • Automatic greenhouses (ensures air is correct temperature for plants)
Light / Lamp	• Security lights • Automatic greenhouse (to give plants correct amount of light)

Examples:



Motors in washing machines make the drum spin.

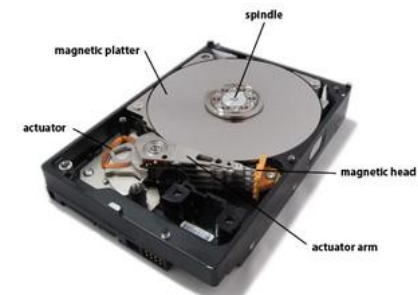
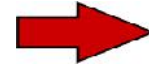
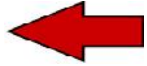


Image of a hard disk drive showing the actuator that makes the disk spin (click to zoom).



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[Section 4: Computer Networks](#)

[Section 7: The Ways in which ICT is used](#)

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